Unit 8, Lesson 5: Finding the Missing Dimension of a Composite Figure

Benchmark

MA.6.GR.2.2 Solve mathematical and real-world problems involving the area of quadrilaterals and composite figures by decomposing them into triangles or rectangles.

Learning Target

- [ ] I can find the value of a missing dimension for a quadrilateral or a composite figure.

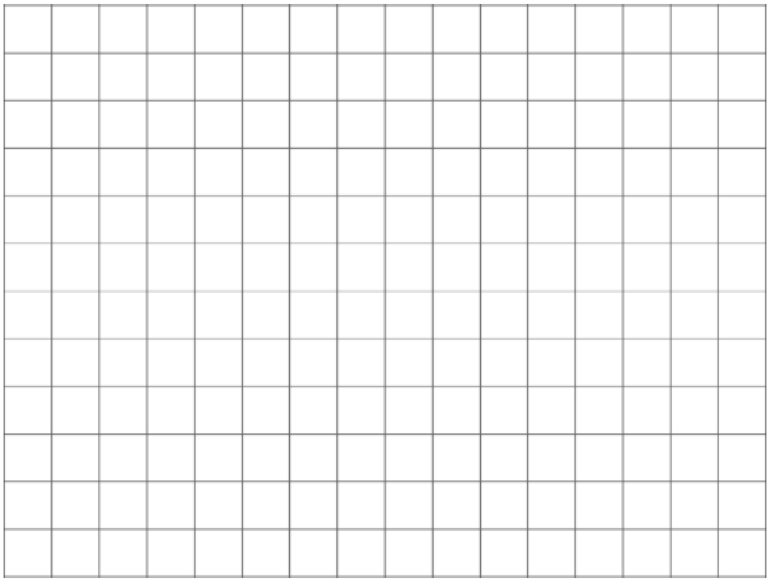
8.5.1 Warm-Up: Closest to Six

1. Use the digits 0 to 9, at most one time each, to fill in ordered pairs for all three points, such that the area of Triangle ABC is closest to 6 square units.

A ($\boxed{}$, $\boxed{}$)

B ($\boxed{}$, $\boxed{}$)

C ($\boxed{}$, $\boxed{}$)

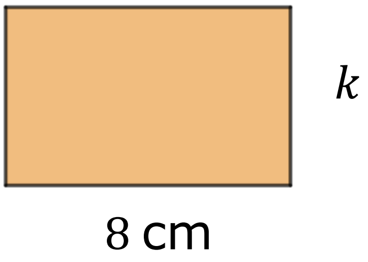
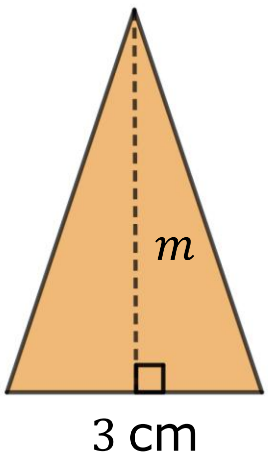
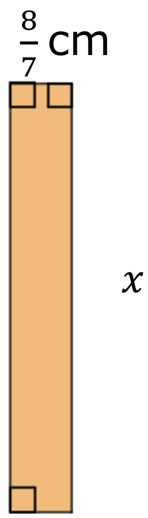


8.5.2 Collaboration: Finding a Missing Value

1. Complete the table by filling in the blanks for the following math facts.

If...	Then...
$40 \div 8 = \_\_\_\_$	$8 \times \_\_\_\_ = 40$
$27 \div 3 \div \frac{1}{2} = \_\_\_\_$	$\frac{1}{2} \times \_\_\_\_ \times 3 = 27$
$8 \div \frac{8}{7} = \_\_\_\_$	$\_\_\_\_ \times \frac{8}{7} = 8$

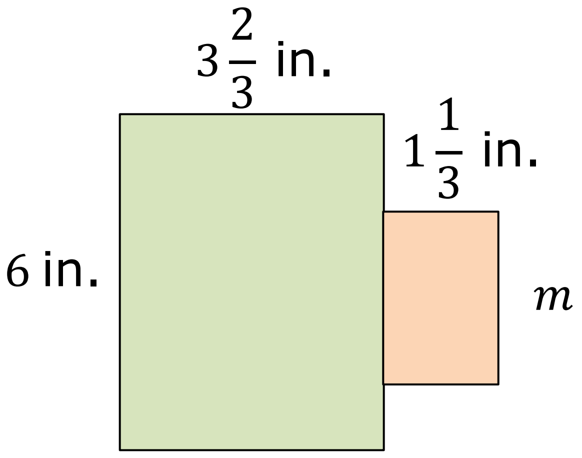
2. With your partner, use the previous math facts to find the missing dimensions of the quadrilaterals. All measurements are in centimeters (cm).

		
Area $= 40$ cm^2	Area $= 27$ cm^2	Area $= 8$ cm^2
$k = \_\_\_\_\_$	$m = \_\_\_\_\_$	$x = \_\_\_\_\_$

8.5.3 Guided Instruction: Finding a Missing Dimension

To find the area of a composite figure, find the sum of the decomposed parts of the figure.

The figure shown is composed of two rectangles.



1. Label the rectangle on the left A, and the rectangle on the right B.

The area of the figure is 26 square inches (in^2).

2. Complete the statements.

The area of the composite figure is the ○ sum ○ difference of the area of Rectangle A and the area of Rectangle B.

Therefore, the area of the composite figure ○ plus ○ minus the area of Rectangle A is equal to the area of Rectangle B.

3. The area of the composite figure is _________ in^2.

4. The area of Rectangle A is ________ in^2.

5. Find the area of Rectangle B.

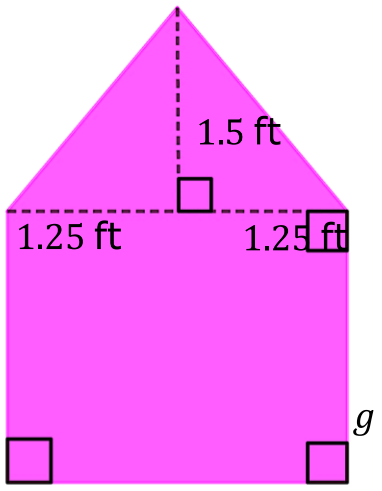
6. Complete the statement.

The missing dimension of Rectangle B can be determined by [○ multiplying ○ dividing] the [○ area ○ height] of Rectangle B by the given dimension.

7. What is the missing dimension, m, of Rectangle B?

8.5.4 Guided Practice: Finding the Value of the Missing Dimension

1. Xavier is painting the back wall of a dollhouse. He knows the back wall of the dollhouse has an area of 6.875 square feet (ft^2).

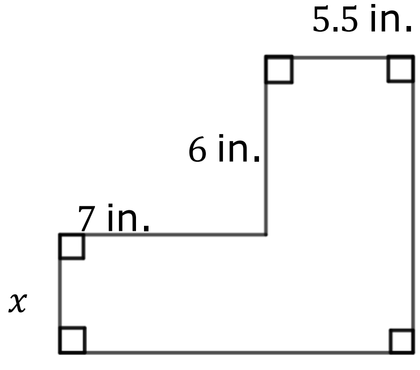


   a. What is the area of the triangular top? Include units.

   b. What is the area of the rectangular base? Include units.

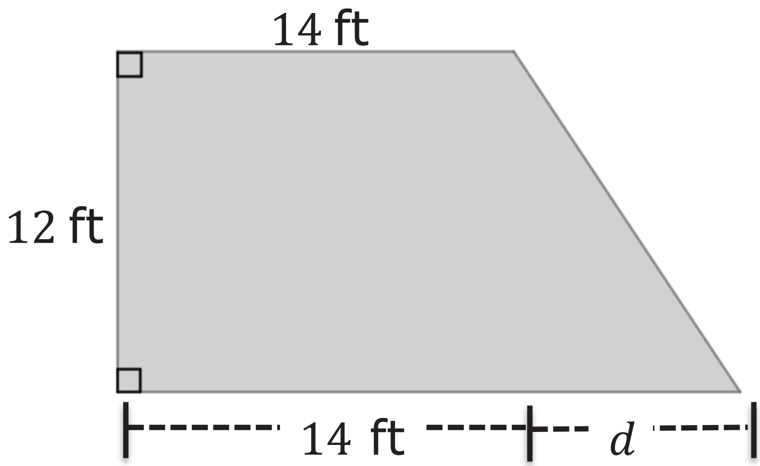
   c. What is the height, g, of the rectangular base? Include units.

2. The area of the composite figure is 84 square inches (in^2). Find the missing dimension. Include units.

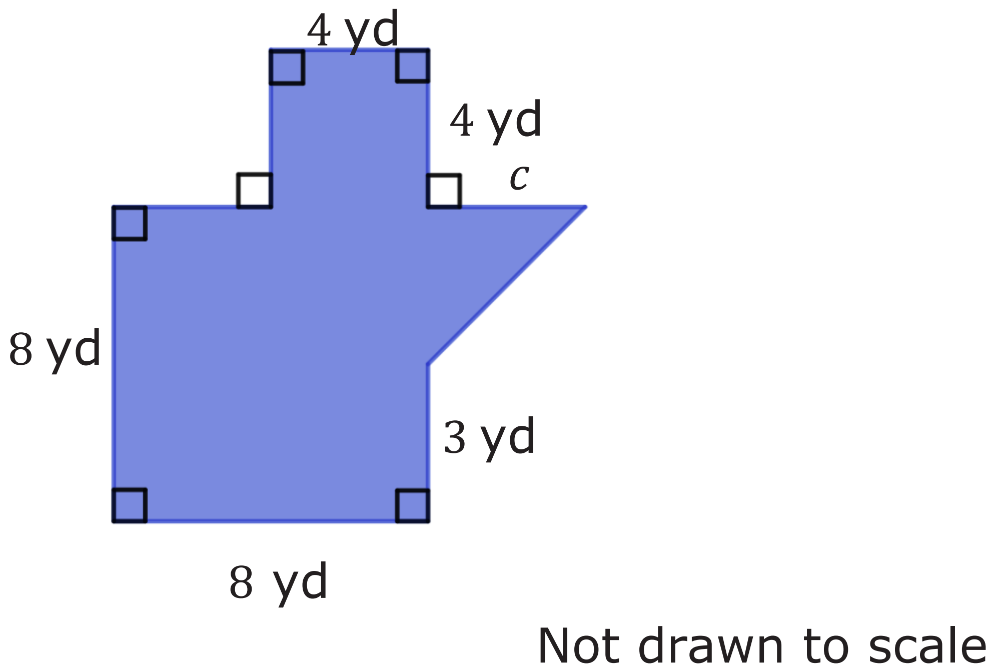


8.5.5 Your Turn!

1. Julio is building a new patio at his house. He needs 216 square feet (ft^2) of material to cover the area of his design, as shown. Find the missing dimension. Include units.



2. The area of the composite figure is 87.5 square yards (yd^2). Find the value of the missing dimension. Include units.



8.5.6 Wrap-Up: Reflection

Complete each statement based about today's lesson.

1. I am still unsure about …

2. One question I still have about today's lesson is …